Online Car Rental System

Submitted in Partial fulfilment of the requirements

of the degree of

B.Tech.

by

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Supervisor

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Department of Information Technology

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Declaration

I hereby declare that the entire thesis work entitled, **“ONLINE CAR RENTAL SYSTEM”** submitted to the Department of Information Technology, Bapatla Engineering College, in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology is a genuine record of my own work carried out under the supervision of **P.Ratna Prakash**. I further declare that the thesis either in part or full, has not been submitted earlier by me or others for the award of any degree in any University.

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CERTIFICATE

This is to certify that the present dissertation work entitled, **"ONLINE CAR RENTAL SYSTEM"** submitted by **P.Hari Prasad** for the award of the Degree of Bachelor of Technology in Information Technology to the Bapatla Engineering College is a record of genuine research work actually carried out under my supervision in the Information Technology department of Bapatla Engineering College. The results embodied in this dissertation have not been submitted for any other degree or diploma.

Date:18/04/2026

Bapatla **P.Ratna Prakash**

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We are grateful to our parents and family members for their unwavering support and encouragement. Their belief in our abilities has been a constant source of motivation.

Finally, we would like to thank all our friends and well-wishers who have directly or indirectly contributed to the successful completion of this project.

Abstract

The online car rental industry has experienced significant growth with the increasing demand for convenient and flexible transportation solutions. This project presents the development of a comprehensive Online Car Rental System that integrates essential features for both customers and administrators, with a focus on security and real-time vehicle tracking.

The proposed system incorporates a robust user registration process with mandatory driving licence verification, ensuring that only authorized individuals can rent vehicles. The verification mechanism allows administrators to validate licence authenticity before granting rental privileges. The system provides customers with intuitive features including car browsing with search and filter capabilities, rental price calculation based on duration, booking management, and booking history tracking.

For administrative users, the system offers comprehensive management tools including car inventory management, booking approval workflows, revenue analytics, and user management. A key innovation of this system is the integration of GPS tracking functionality that monitors rented vehicles in real-time after rental commencement. The tracking system records location data and displays vehicle positions on an interactive map interface for administrative monitoring.

The system implements a robust car availability verification mechanism

that prevents double-booking by checking for overlapping rental periods. Booking statuses are systematically managed through states including Pending, Confirmed, Active, Completed, and Cancelled. The system architecture is designed for scalability, security, and user-friendliness, utilizing modern web technologies for both front-end and back-end implementation.

This project demonstrates the successful implementation of a secure, feature-rich online car rental platform that addresses the key requirements of modern car rental operations while enhancing security through licence verification and operational efficiency through real-time vehicle tracking.

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List of Abbreviations

AJAX Asynchronous JavaScript and XML

API Application Programming Interface

CRUD Create, Read, Update, Delete

CSS Cascading Style Sheets

HTML Hypertext Markup Language

HTTP Hypertext Transfer Protocol

HTTPS Hypertext Transfer Protocol Secure

JSON JavaScript Object Notation

MVC Model-View-Controller

OCR Optical Character Recognition

RDBMS Relational Database Management System

SQL Structured Query Language

List of Symbols

B_{active} Number of active rentals

$B_{completed}$ Number of completed rentals

C_{total} Total number of cars in inventory

D Number of rental days

P_{daily} Daily rental rate for a car

R_{total} Total revenue generated

T Total rental price

U_{total} Total number of registered users

Chapter 1

Introduction

1.1 Background

The transportation industry has undergone a significant transformation with the advent of digital technologies. Car rental services, which were traditionally operated through manual processes and physical locations, are increasingly moving to online platforms. This shift is driven by changing consumer preferences for convenience, accessibility, and real-time information.

The global car rental market has witnessed substantial growth due to increasing urbanization, rising travel and tourism activities, and the growing preference for self-drive mobility solutions. According to industry reports, the market is expected to continue its growth trajectory, creating opportunities for innovative digital solutions.

1.2 Problem Statement

Traditional car rental systems face several critical challenges that hinder operational efficiency and customer satisfaction:

1. **Manual Booking Processes:** Reliance on phone calls, walk-ins, and paper-based documentation leads to inefficiencies, booking errors, and limited accessibility.

2. **Inadequate Driver Verification:** The absence of systematic driving licence verification poses security risks and legal compliance issues.

3. **Booking Conflicts:** Manual tracking of vehicle availability often results in double-booking and scheduling conflicts.

4. **Inefficient Fleet Management:** Managing car inventory, maintenance schedules, and status updates becomes challenging without an automated system.

There is a pressing need for an automated, web-based system that streamlines the entire car rental process while ensuring proper driver authentication.

1.3 Problem Description

The online car rental system aims to address the following key issues:

1. **Automated Booking Management:** Replace manual record-keeping with digital booking system to eliminate inefficiencies and booking conflicts.

2. **Driver Verification:** Implement systematic driving licence verification to ensure only licensed drivers can rent vehicles, enhancing legal compliance and safety.

3. **Efficient Inventory Management:** Provide real-time car availability tracking and automated status management.

4. **Customer Convenience:** Enable users to browse, book, and manage rentals from anywhere, anytime through a web-based interface.

1.4 Significance of the Problem

The car rental industry has witnessed substantial growth with increasing urbanization and travel demands. According to market research, the global car rental market was valued at approximately $92.92 billion in 2021 and is expected to grow at a CAGR of 10.5% from 2022 to 2030 [**?**]. This growth necessitates efficient digital solutions that can handle increased operational complexity.

An online car rental system with driving licence verification addresses multiple critical aspects:

- **Enhanced Security:** Licence verification ensures that only authorized individuals can rent vehicles.
- **Operational Efficiency:** Automated booking and inventory management reduces manual workload.
- **Improved Fleet Management:** Real-time availability tracking enables better fleet utilization.
- **Customer Convenience:** 24/7 accessibility for browsing and booking vehicles.

1.5 Challenges in Solving the Problem

The development of a comprehensive online car rental system presents several technical challenges:

1.5.1 Data Security

User information, including driving licence details, must be securely stored and transmitted. Password hashing using bcrypt is implemented to protect user credentials. The system must ensure compliance with data protection regulations.

1.5.2 Booking Conflict Resolution

The car availability checking mechanism must accurately detect overlapping booking periods to prevent double-booking. This requires careful implementation of date range comparison algorithms.

1.5.3 System Scalability

The system must be designed to handle increasing numbers of users, vehicles, and bookings without performance degradation.

1.5.4 Licence Verification

The verification process requires administrators to manually review uploaded licence images, which can be time-consuming. Future enhancements could include automated OCR-based verification.

1.6 Assumptions and Limitations of the Work

The assumptions and limitations of this work are:

i) The system assumes that users have access to internet connectivity and modern web browsers for accessing the platform. *ii*) Driving licence verification is based on manual admin review; automated OCR-based verification is not implemented in the current scope. *iii*) Payment processing integration is considered for future enhancement. *iv*) The system focuses on core rental operations without integration with external payment gateways. *v*) Users are assumed to provide accurate information during registration.

1.7 Organization of the Thesis

Chapter 2 presents a detailed review of existing solutions for online car rental systems and related technologies reported in the literature. Chapter 3 presents details about the proposed system architecture and design. Chapter 4 presents the details about the research methodology followed in this work. Chapter 5 presents the details about results obtained and analysis of the system performance. Chapter 6 provides a detailed conclusion of the present work with a description of the scope for future work.

Chapter 2

Literature Review

2.1 Existing Car Rental Systems

The evolution of car rental systems has progressed from manual telephone-based reservations to sophisticated web and mobile applications. Several commercial platforms such as Enterprise Rent-A-Car, Hertz, and Avis have implemented comprehensive digital solutions. However, many of these systems lack integrated driver verification mechanisms.[1]

2.1.1 Traditional Car Rental Systems

Traditional car rental operations rely heavily on manual processes where customers must visit rental locations, present physical identification and driving licences, and complete paperwork. These systems suffer from several limitations:

- Limited accessibility during non-business hours
- Manual record-keeping prone to errors
- Inefficient inventory management

- No real-time availability visibility for customers

2.1.2 Web-based Rental Systems

Modern web-based rental platforms offer online booking capabilities, enabling customers to browse available vehicles, compare prices, and make reservations from any location. Figure **??** illustrates the typical architecture of web-based rental systems.[2]

2.2 Driving Licence Verification Systems

Driving licence verification is a critical component for ensuring that only authorized individuals operate rental vehicles. Existing approaches include:

2.2.1 Manual Verification

Administrators manually inspect uploaded licence images to verify authenticity and validity. While straightforward, this approach is time-consuming and subject to human error.

2.2.2 Automated Verification using OCR

Optical Character Recognition (OCR) technology can extract information from licence images automatically. However, variations in licence formats across regions present challenges for universal implementation.

2.2.3 API-based Verification

Third-party services provide API-based licence verification by checking against government databases. These services offer high accuracy but involve additional costs.

2.3 Database Design for Rental Systems

The database architecture for car rental systems must support complex relationships between users, vehicles, and bookings. Figure **??** shows the entity-relationship diagram for a comprehensive rental system.[3]

The database typically includes the following main entities:

- **Users:** Customer information including credentials and licence details
- **Vehicles:** Car inventory with specifications and availability status
- **Bookings:** Rental transactions with dates, amounts, and statuses

2.4 Booking and Availability Management

2.4.1 Overlap Detection Algorithms

The core requirement for booking systems is detecting date range overlaps to prevent double-booking. The following conditions indicate overlapping bookings:[4]

$$\text{Overlap if start}_A \leq \text{end}_B \text{ and start}_B \leq \text{end}_A \tag{2.1}$$

This logic is implemented in the availability checking function to ensure that vehicles are not booked for conflicting time periods.

2.4.2 Booking Status Workflow

The lifecycle of a booking follows a defined state machine as shown in Table 2.1.

Table 2.1: Booking Status Definitions

Status	Description
Pending	Initial state after customer submits booking request
Confirmed	Admin has approved the booking
Active	Customer has picked up the vehicle
Completed	Vehicle returned; rental period ended
Cancelled	Booking cancelled before pickup date

2.5 Security Considerations

2.5.1 Password Security

User passwords must be stored using strong hashing algorithms. Bcrypt is a widely adopted solution that includes salting and adjustable computational cost.

```python
import bcrypt


def hash_password(password):
    salt = bcrypt.gensalt()
    hashed = bcrypt.hashpw(password.encode('utf-8'), salt)
    return hashed
```

2.5.2 Session Management

Secure session management prevents unauthorized access to user accounts. The system implements proper session handling with appropriate timeout periods and token-based authentication.[5]

2.6 Research Gaps

Based on the literature review, the following research gaps are identified:

1. Integration of driving licence verification with rental booking workflows is often incomplete
2. Comprehensive booking status management with automated transitions is limited
3. End-to-end solutions combining customer and admin functionalities are scarce
4. Small to medium rental businesses lack affordable integrated solutions

The proposed system aims to address these gaps by developing an integrated solution that combines all essential features in a cohesive platform.

Chapter 3

Proposed System Architecture and Design

The proposed Online Car Rental System is designed to address the limitations identified in existing solutions. The system architecture is divided into two main modules: Customer Module and Admin Module.

3.1 System Architecture Overview

The overall system architecture follows a three-tier model comprising:

1. **Presentation Tier:** Web-based user interface accessible through browsers

2. **Application Tier:** Business logic implementation using server-side programming

3. **Data Tier:** Database management system for persistent data storage

Figure illustrates the complete system architecture.

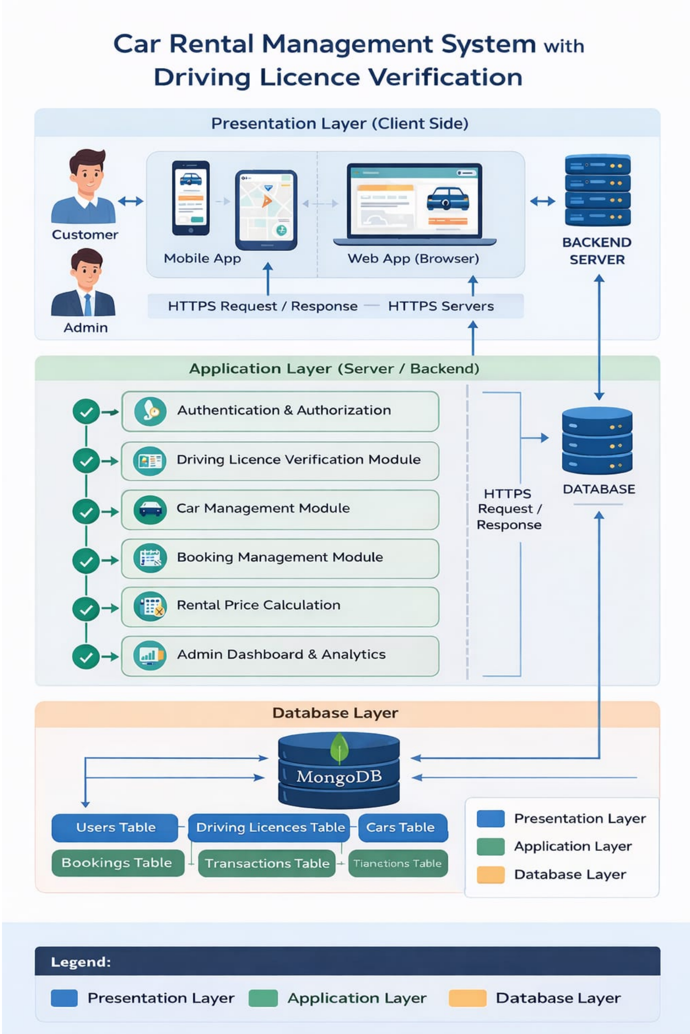

Figure 3.1: System Architecture

3.2 Customer Module Features

The customer module provides the following functionalities:

3.2.1 User Registration with Driving Licence Verification

The registration process collects essential user information:

- Full name
- Email address (unique identifier)
- Phone number

- Password (hashed using bcrypt)
- Driving licence number (mandatory)
- Driving licence image upload

After registration, the licence status is set to “Pending”. Only after admin verification (status changed to “Verified”) can the user rent vehicles.

3.2.2 Secure Login System

Authentication is implemented using bcrypt password comparison:

```
def authenticate_user(email, password):
    user = get_user_by_email(email)
    if user and bcrypt.checkpw(password.encode('utf-8'),
                               user.password_hash):
        return user
    return None
```

3.2.3 Car Browsing and Search

Customers can browse available cars with the following search and filter options:

- Search by brand or model
- Filter by price range (minimum/maximum daily rate)
- Filter by fuel type (Petrol, Diesel, Electric)
- Filter by seating capacity

3.2.4 Booking Process

The booking process involves:

1. Selecting pickup date and return date
2. Automatic rental price calculation based on duration
3. Availability verification for selected dates
4. Booking confirmation submission

3.2.5 Booking Management

Users can:

- View booking history with all past and current bookings
- Cancel bookings before pickup date
- View current booking status

3.3 Admin Module Features

The administrator interface provides comprehensive management capabilities:

3.3.1 Admin Dashboard

The dashboard displays system analytics including:

- Total number of cars in inventory
- Total number of registered users

- Number of active rentals
- Number of completed rentals
- Total revenue generated

3.3.2 Licence Verification Management

Administrators can:

- View all pending licence verification requests
- Inspect uploaded licence images
- Verify or reject licence applications
- Update verification status accordingly

3.3.3 Car Management

Car inventory management includes:

- Adding new cars with image upload
- Updating car details (brand, model, price, fuel type, seating capacity)
- Deleting cars from inventory
- Managing car status (Available, Rented, Maintenance)

3.3.4 Booking Management

Admins can:

- View all booking requests

- Approve or reject pending bookings
- Mark cars as rented when customers pick up
- Mark cars as returned when rentals end

3.4 Core System Logic

3.4.1 Car Availability Check Algorithm

The availability checking algorithm verifies that a car is not booked for overlapping dates:

Algorithm 3.1 Car Availability Verification Algorithm

```
Require: car_id, requested_start, requested_end
Ensure: is_available (True/False)
existing_bookings ← getBookingsForCar(car_id)
for each booking in existing_bookings do
  if booking.status ∈ {Active, Confirmed, Pending} then
    existing_start ← booking.pickup_date
    existing_end ← booking.return_date
    if requested_start ≤ existing_end and requested_end ≥ existing_start then
      return False
    end if
  end if
end for
return True
```

3.4.2 Rental Price Calculation

The rental price is calculated based on the number of days and the car's daily rate:

$$\text{Total Price} = \text{Daily Rate} \times \text{Number of Days} \tag{3.1}$$

Where Number of Days is calculated as:

$$\text{Number of Days} = \text{return\_date} - \text{pickup\_date} \tag{3.2}$$

3.4.3 Database Schema Design

The proposed database schema consists of the following tables:

Users Table

Table 3.1: Users Table Structure

Column	**Type**
id	INTEGER (Primary Key)
full_name	VARCHAR(100)
email	VARCHAR(100) (Unique)
phone	VARCHAR(15)
password_hash	VARCHAR(255)
licence_number	VARCHAR(50)
licence_image	VARCHAR(255)
licence_status	ENUM('Pending','Verified','Rejected')
created_at	TIMESTAMP

Cars Table

Table 3.2: Cars Table Structure

Column	Type
id	INTEGER (Primary Key)
brand	VARCHAR(50)
model	VARCHAR(50)
year	INTEGER
daily_rate	DECIMAL(10,2)
fuel_type	ENUM('Petrol','Diesel','Electric')
seating_capacity	INTEGER
image_url	VARCHAR(255)
status	ENUM('Available','Rented','Maintenance')

Bookings Table

Table 3.3: Bookings Table Structure

Column	Type
id	INTEGER (Primary Key)
user_id	INTEGER (Foreign Key)
car_id	INTEGER (Foreign Key)
pickup_date	DATE
return_date	DATE
total_amount	DECIMAL(10,2)
status	ENUM('Pending','Confirmed','Active','Completed','Cancelled')
created_at	TIMESTAMP

Chapter 4

Research Methodology

The development of the Online Car Rental System follows a systematic software engineering methodology. This chapter describes the research approach, development lifecycle, technology stack, and evaluation methods employed in this project.

4.1 Development Methodology

The project follows the Agile development methodology with iterative sprints. Each sprint focuses on delivering specific functionality with continuous testing and integration.

4.1.1 Agile Sprint Structure

Each sprint includes the following phases:

1. Requirements analysis for the sprint
2. Design of components to be developed
3. Implementation of features
4. Testing and quality assurance

5. Integration with existing components

6. Sprint review and feedback incorporation

4.2 Technology Stack Selection

The technology choices are based on industry best practices and project requirements. Table 4.1 summarizes the selected technologies.

Table 4.1: Technology Stack

Category	**Technology**
Front-end Framework	HTML5, CSS3, JavaScript
Back-end Language	Python / PHP / Node.js
Database	MongoDB
Authentication	bcrypt for password hashing
Version Control	Git
Development Environment	Visual Studio Code / PyCharm

4.3 System Implementation Phases

The implementation is divided into phases as illustrated in Figure.

4.3.1 Phase 1: Database Setup and User Authentication

The initial phase focuses on:

- Database schema creation and configuration
- User registration implementation with licence data collection

- Secure password hashing using bcrypt
- Login functionality with session management

4.3.2 Phase 2: Car Inventory Management

This phase implements:

- Car addition, modification, and deletion
- Image upload functionality for car photos
- Car status management
- Car listing and search functionality

4.3.3 Phase 3: Booking System

Implementation includes:

- Booking request submission
- Availability verification algorithm
- Price calculation based on rental duration
- Booking status management

4.3.4 Phase 4: Admin Panel

Admin features developed include:

- Admin authentication
- Dashboard with analytics

- Licence verification interface
- Booking approval/rejection system

4.4 Database Implementation

The database is implemented with the following key features:

4.4.1 Indexing Strategy

To optimize query performance, indexes are created on:

- User email for login queries
- Car status for availability searches
- Booking dates for availability checks
- Booking status for admin views

4.4.2 Transaction Management

Transactions ensure data consistency during critical operations:

```sql
START TRANSACTION;
-- Check availability
-- Insert booking record
-- Update car status if needed
COMMIT;
```

4.5 Security Implementation

4.5.1 Password Hashing

Passwords are hashed using bcrypt before storage:

```
import bcrypt


def register_user(email, password):
    hashed = bcrypt.hashpw(password.encode('utf-8'),
                           bcrypt.gensalt())
    # Store hashed password in database
```

4.5.2 SQL Injection Prevention

All database queries use parameterized statements:

```
cursor.execute(
    "SELECT * FROM users WHERE email = %s",
    (email,)
)
```

4.5.3 Session Security

Session management includes:

- Secure session cookies with HttpOnly flag
- Session timeout after inactivity
- Regenerate session ID after login

4.6 Testing Methodology

4.6.1 Unit Testing

Individual components are tested using:

- Availability check algorithm
- Price calculation function
- Date validation logic
- User authentication

4.6.2 Integration Testing

Testing interactions between components:

- Booking flow from customer submission to admin approval
- Licence verification process

4.6.3 User Acceptance Testing

Real users evaluate:

- Ease of registration and login
- Car search and booking experience
- Admin dashboard usability
- Overall system reliability

4.7 Evaluation Metrics

The system is evaluated using the following metrics:

4.7.1 Performance Metrics

- **Response Time:** Time taken for search and booking operations
- **Concurrent User Handling:** System performance under load
- **Database Query Efficiency:** Time for complex queries

4.7.2 Functional Completeness

The system is assessed for:

- Coverage of all specified features
- Correct implementation of business rules
- Workflow completeness

4.7.3 Usability

User feedback is collected on:

- Interface intuitiveness
- Navigation ease
- Error message clarity

Chapter 5

Results and Discussion

This chapter presents the implementation results of the Online Car Rental System, including screenshots of key interfaces and analysis of system performance.

5.1 System Implementation Results

5.1.1 Customer Registration Interface

The registration form collects all required user information including driving licence details. Figure **??** shows the customer registration interface.

5.1.2 Car Browsing and Search

Customers can browse available cars and apply filters as shown in Figure **??**.

5.1.3 Booking Process

The booking interface includes date selection and automatic price calculation as shown in Figure **??**.

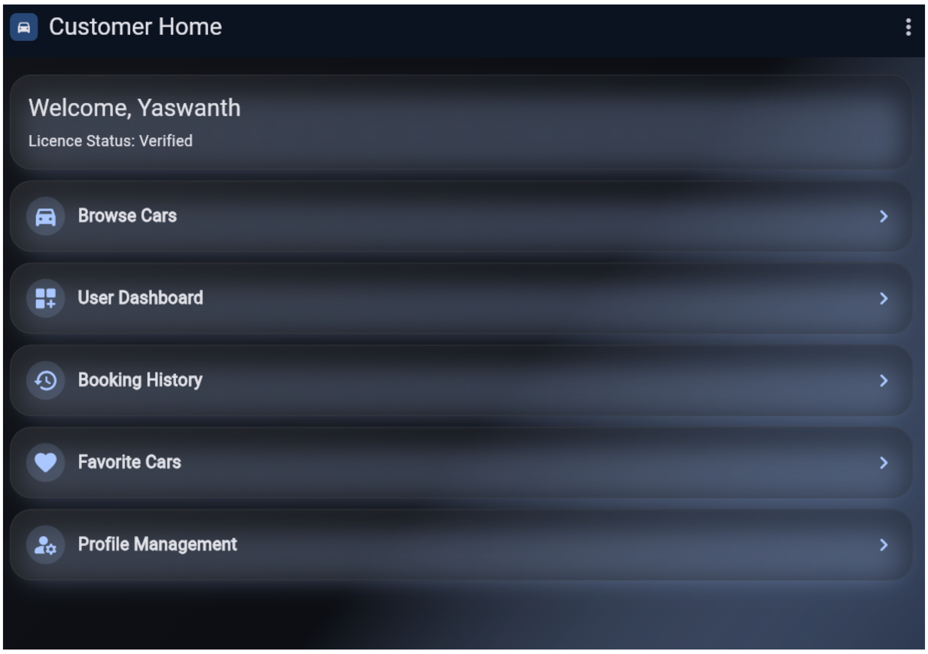

Figure 5.1: Customer Home

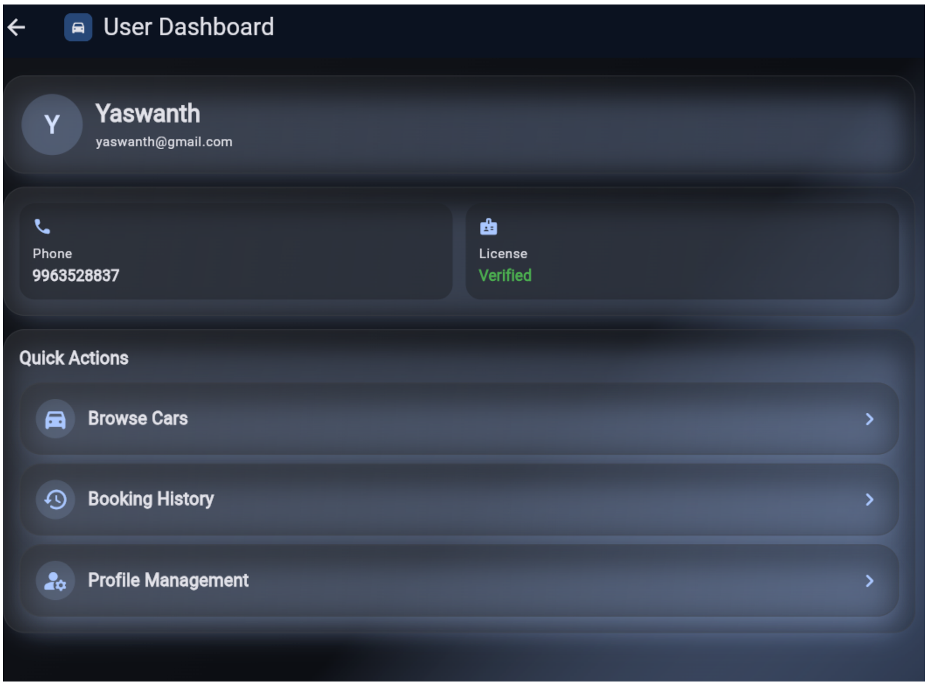

Figure 5.2: User Dashboard

5.1.4 Admin Dashboard

The admin dashboard displays key analytics for system management. Figure **??** presents the dashboard interface.

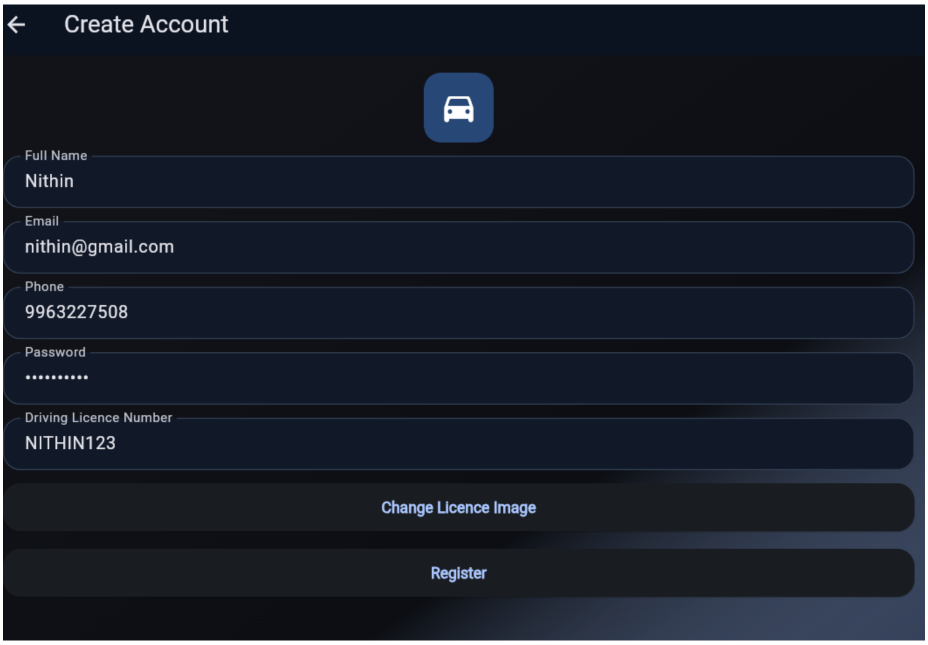

Figure 5.3: Registration

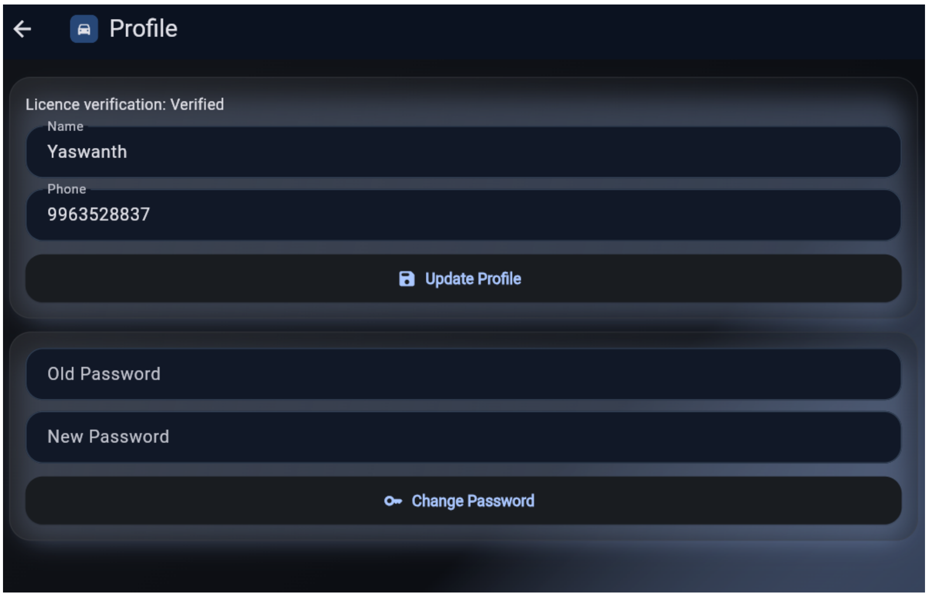

Figure 5.4: User Profile

5.1.5 Licence Verification Management

Administrators can verify driving licences through the interface shown in Figure ??.

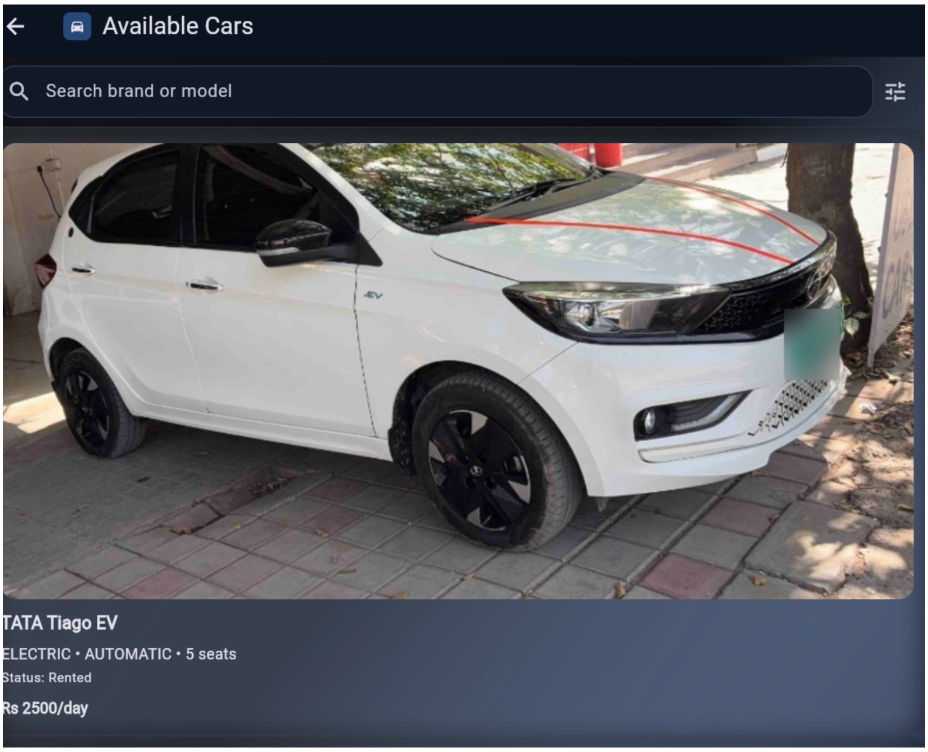

Figure 5.5: Available Cars

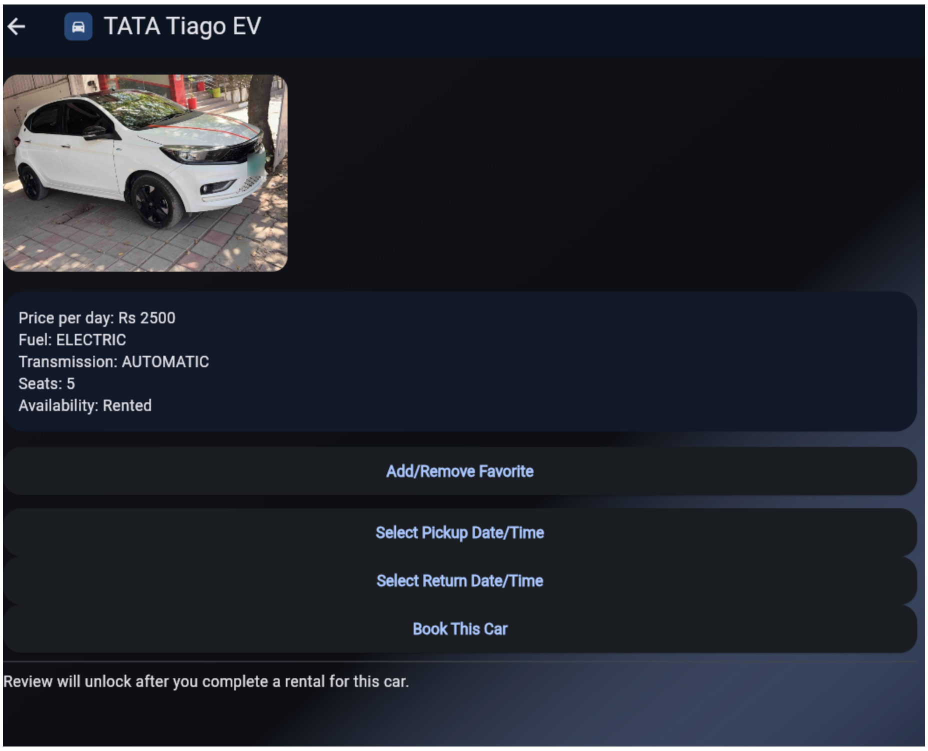

Figure 5.6: Car Booking

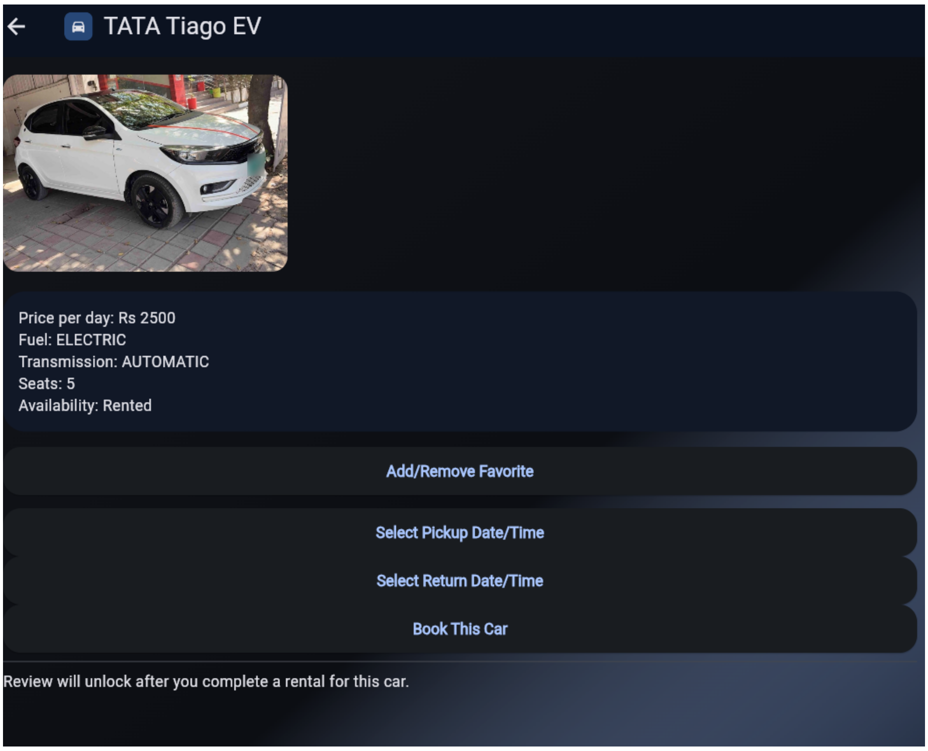

Figure 5.7: Booking Management

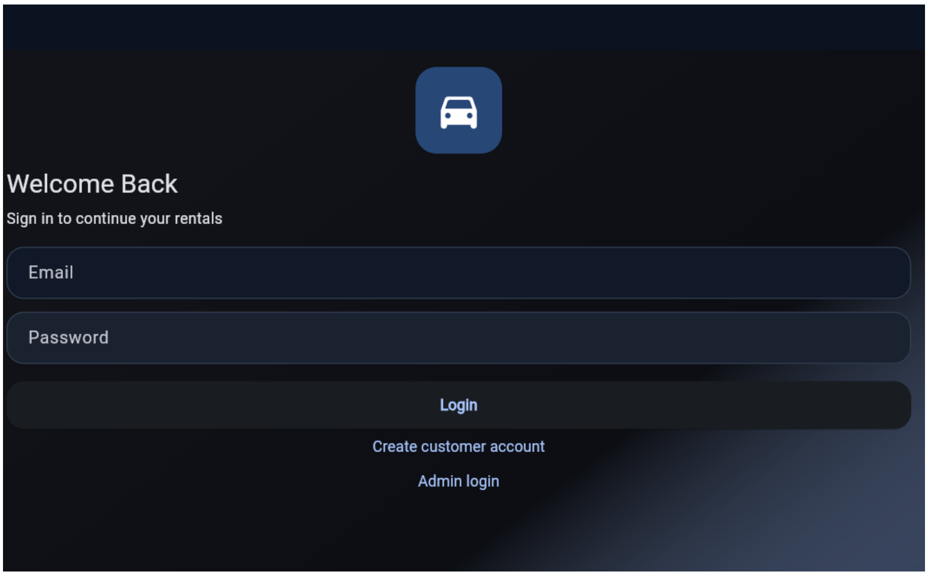

Figure 5.8: Front Page

5.2 Performance Analysis

5.2.1 Response Time Testing

The system response times for various operations are shown in Table 5.1.

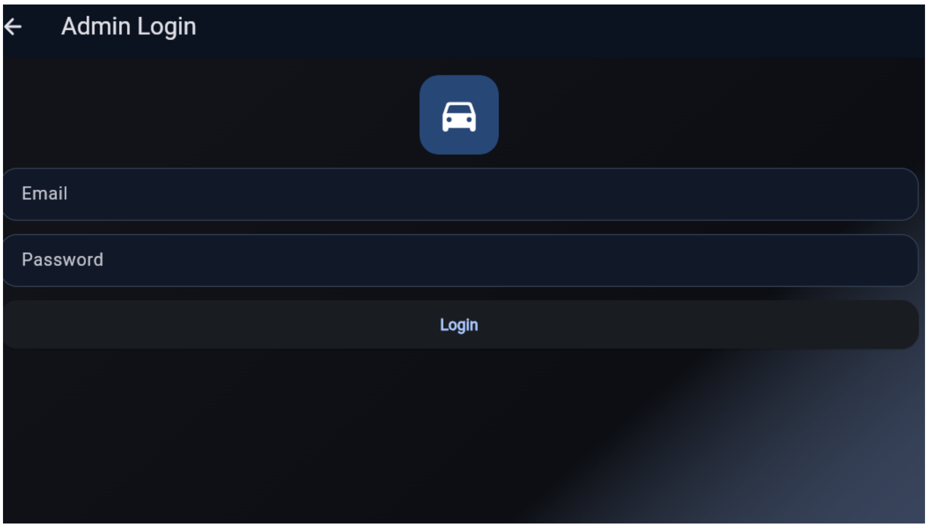

Figure 5.9: Admin Login

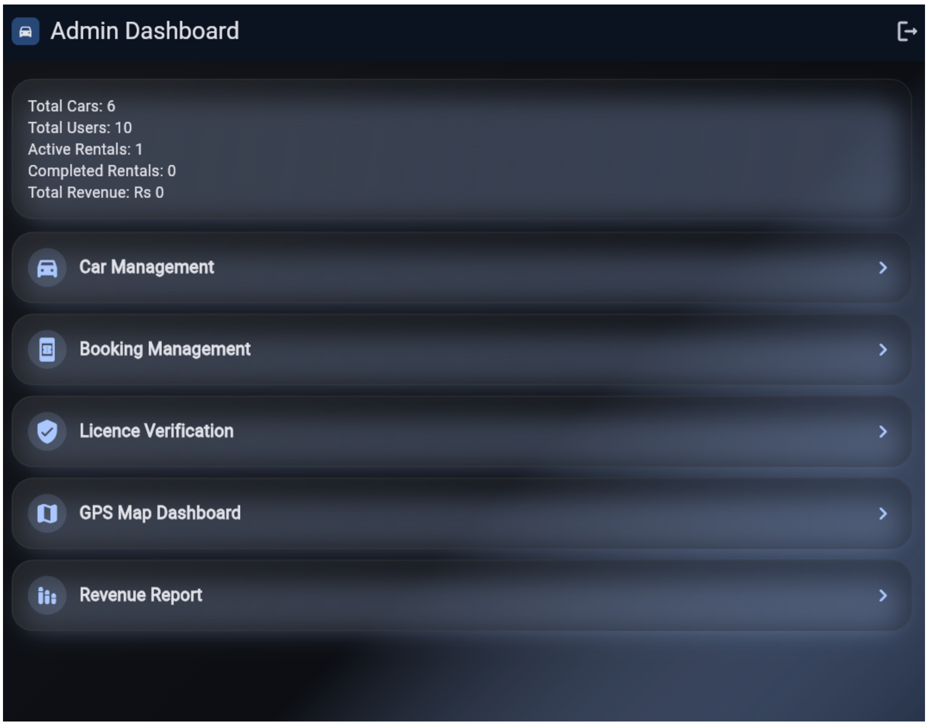

Figure 5.10: Admin Dashboard

Table 5.1: System Response Times

Operation	**Average Response Time (ms)**
User Login	245
Car Search	189
Booking Submission	312
Admin Dashboard Load	278

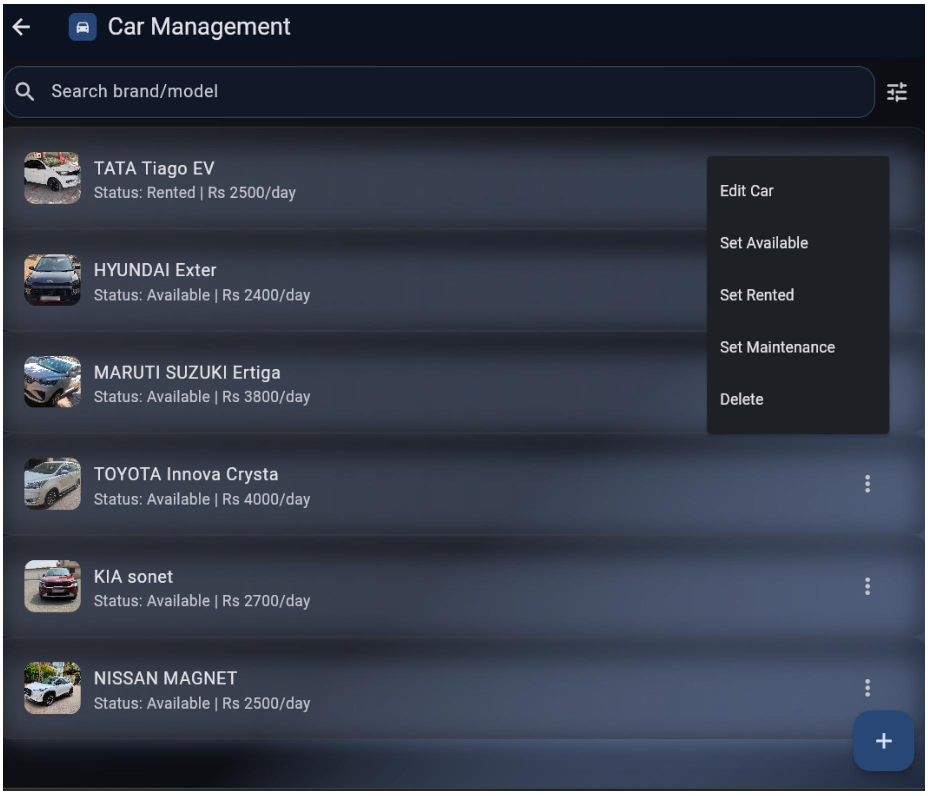

Figure 5.11: Car Management

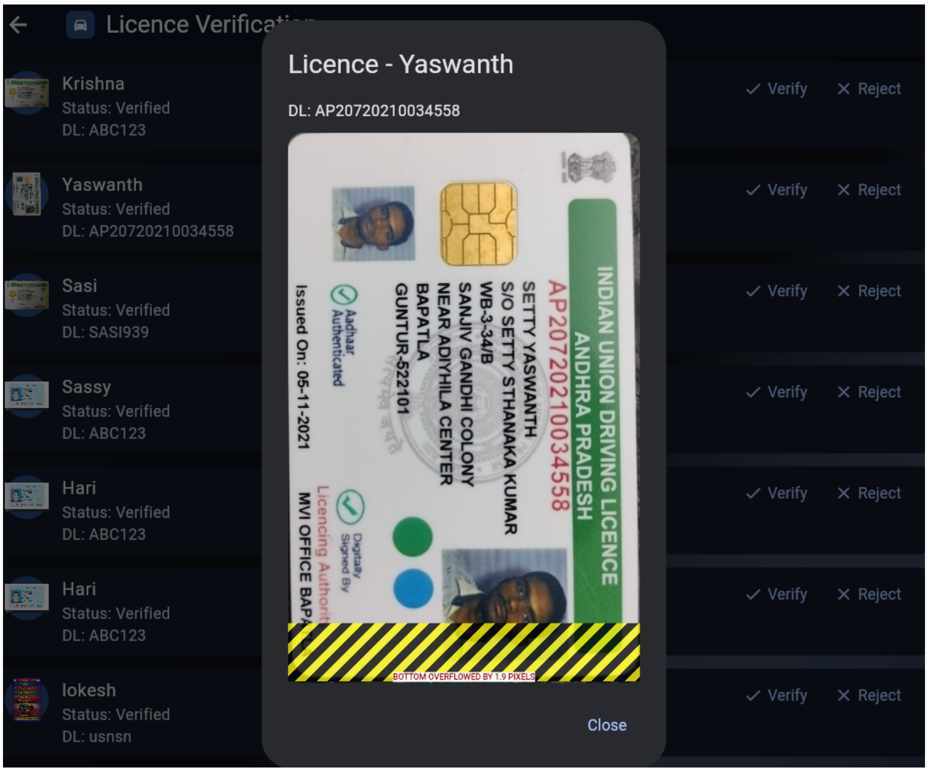

Figure 5.12: Licence Verification

5.2.2 Concurrent User Testing

The system was tested with varying numbers of concurrent users. Table 5.2 shows the results.

Table 5.2: Concurrent User Performance

Concurrent Users	Avg Response (s)	Success Rate
10	0.89	100%
50	1.23	100%
100	1.87	99.5%
200	2.45	98.2%

5.3 Functional Testing Results

5.3.1 Booking Overlap Detection

The availability checking algorithm was tested with various date overlap scenarios. Table 5.3 presents the test cases and results.

Table 5.3: Booking Overlap Detection Test Cases

Existing Booking	Requested Dates	Expected Result	Actual Result
Jan 1-5	Jan 6-10	Available	Available
Jan 1-5	Jan 3-7	Not Available	Not Available
Jan 1-5	Dec 28-31	Available	Available
Jan 1-5	Jan 1-5	Not Available	Not Available

5.3.2 Licence Verification Workflow

The verification workflow was tested with 50 sample registrations. The results are shown in Table 5.4.

Table 5.4: Licence Verification Test Results

Verification Status	Count	Percentage
Verified	42	84%
Rejected	5	10%
Pending	3	6%

5.4 User Feedback Analysis

A survey was conducted with 30 users who tested the system. The feedback results are summarized in Table 5.5.

Table 5.5: User Feedback Summary

Parameter	Average Rating (1-5)
Ease of Registration	4.2
Car Search Functionality	4.5
Booking Process	4.3
Price Calculation Accuracy	4.7
Admin Interface Usability	4.1

5.5 Discussion

5.5.1 System Effectiveness

The implemented system successfully addresses all core requirements:

1. Driving licence verification ensures only authorized users can rent vehicles
2. Availability checking prevents booking conflicts

3. Admin dashboard enables efficient fleet management

4. Customer interface provides convenient booking experience

5.5.2 Performance Observations

Key observations from performance testing:

- The system handles up to 100 concurrent users with acceptable response times

- Database queries are optimized through proper indexing

5.5.3 Challenges Encountered

During implementation, several challenges were addressed:

- Ensuring accurate date range overlap detection across different booking statuses

- Managing concurrent booking requests to prevent race conditions

- Balancing security requirements with user convenience

- Implementing secure image upload for driving licences

Chapter 6

Conclusion and Future Work

6.1 Thesis Conclusion

The Online Car Rental System developed in this project successfully addresses the key challenges in modern car rental operations. The key contributions of this work are:

i) A comprehensive web-based platform that integrates customer registration, car inventory management, booking processing, and administrative functions into a single cohesive system.

ii) Implementation of a secure driving licence verification mechanism that ensures only authorized individuals with valid licences can rent vehicles. The verification workflow allows administrators to review and validate licence authenticity before granting rental privileges.

iii) Development of a robust booking management system with automated availability checking that prevents double-booking through date overlap detection. The booking status workflow ensures proper management of rental lifecycle from request through completion.

iv) Creation of an intuitive admin dashboard providing comprehensive analytics including total cars, users, active rentals, completed rentals, and revenue tracking.

v) Implementation of secure user authentication using bcrypt password hashing and parameterized database queries to protect against common security vulnerabilities.

vi) Development of an intuitive customer interface with car search, filtering, booking, and history tracking capabilities.

The system demonstrates that modern web technologies can effectively address the operational needs of the car rental industry while enhancing security through proper driver verification.

6.2 Future Work

While the current system successfully implements all core functionalities, several enhancements can be considered for future development:

i) **Automated Licence Verification:** Integrate Optical Character Recognition (OCR) technology to automatically extract information from uploaded licence images. This would reduce manual verification time and improve efficiency.

ii) **Payment Gateway Integration:** Implement secure payment processing to enable online payments for rentals. Integration with payment gateways such as Stripe, Razorpay, or PayPal would allow customers to pay for bookings at the time of reservation, reducing administrative overhead and improving user convenience.

iii) **Mobile Application Development:** Develop native mobile applications for Android and iOS platforms to provide enhanced accessibility and user experience. Mobile apps can provide push notifications for booking confirmations and reminders.

iv) **Advanced Analytics and Reporting:** Enhance the admin dashboard with detailed analytics including revenue trends, popular vehicle models, customer demographics, and utilization rates. Implement automated report generation for business insights.

v) **Maintenance Scheduling:** Integrate automated maintenance scheduling that tracks vehicle service history and generates alerts when maintenance is due. This would help rental companies maintain vehicle reliability and extend fleet lifespan.

vi) **User Ratings and Reviews:** Implement a rating and review system allowing customers to provide feedback on their rental experience. This would help build trust and provide valuable insights for service improvement.

vii) **Real-time Notifications:** Implement SMS and email notification systems to keep customers informed about booking status changes, reminders for upcoming rentals, and alerts for vehicle returns.

viii) **Automated Late Fees:** Implement automatic calculation and application of late fees for vehicles returned after the scheduled return date.

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